

DATA CLEANING PROCESS

Step 1: Appending Data

Since data was scraped city-wise and stored as separate CSV files, all city files were appended (combined) into one master dataset.

Purpose:

- To create one unified dataset for analysis.
 - To perform cleaning in a centralized way.
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Step 2: Removing Blank Records

- Filtered rows where key columns like car_brand or price were null.
- Removed empty rows that occurred during scraping.

Purpose:

- Prevent inaccurate aggregation.
 - Ensure data reliability.
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Step 3: Removing Duplicate Records

Duplicates were removed based on:

- Car Name
- Year
- KM Driven
- City

Purpose:

- Avoid repeated listings affecting price averages.
 - Maintain dataset integrity.
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Step 4: Cleaning Price Column

Original format examples:

- ₹5.99 Lakh

- ₹1.2 Crore

Cleaning steps:

1. Removed ₹ symbol.
2. Split price into:
 - price_value (numeric part)
 - price_unit (Lakh / Crore)
3. Converted price into INR format.

Conversion logic:

If Lakh → multiply by 100000

If Crore → multiply by 10000000

Purpose:

- Enable accurate numerical analysis.
 - Allow calculation of averages, comparisons.
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Step 5: Standardizing Text Columns

- Trimmed extra spaces.
- Ensured consistent fuel naming.
- Ensured transmission values are consistent.

Purpose:

- Prevent grouping errors in visuals.
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Step 6: Creating Derived Columns

Created:

- Price Range
- KM Range
- Car Age (Current Year – Year)

Purpose:

- Improve analytical depth.
- Enable segmentation analysis.